

SATQUERY AI

An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

- NATURAL-LANGUAGE QUERY
- MULTIMODAL ANALYSIS
- AGENTIC AI ORCHESTRATION

Problem Statement ID: 26167

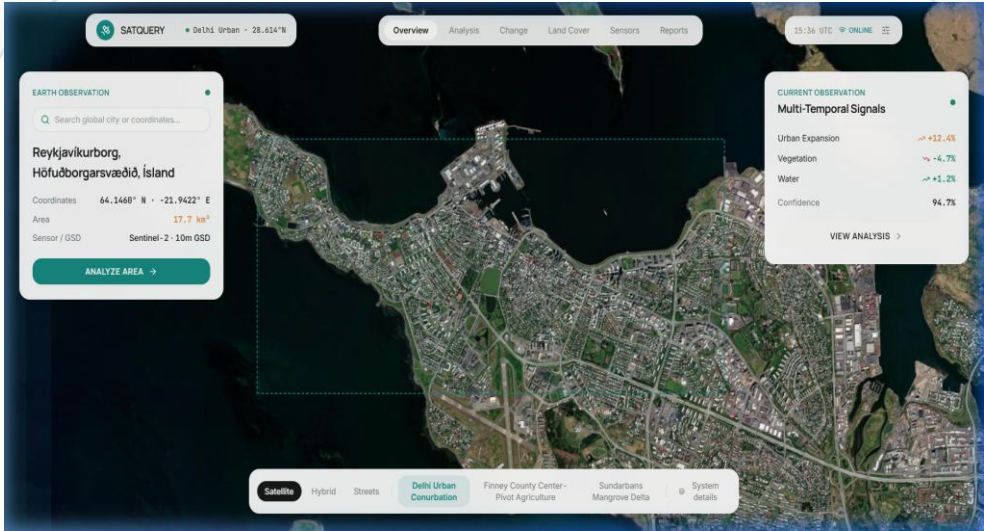
Problem Statement Title: SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

Organization: Indian Space Research Organisation (ISRO)

Department: Department of Space / Indian Space Research Organisation

Theme & Category: Space Technology · Software

Team Name & ID: Team SATQuery · SIH2026-26167



“From Satellite Data → Intelligent Evidence”

Verified Production System · SegFormer · BLIP-2 · CDVQA · Optical + SAR Fusion

SATQUERY AI — FROM COMPLEX GIS TO NATURAL-LANGUAGE AI



FRAGMENTED TOOLS

Isolated specialist workflows

Land cover, change detection, VQA & SAR are siloed in disparate software.



EXPERT DEPENDENCY

High technical barrier

Requires manual parameter tuning: bands, radiometric calibration & CRS transforms.



MULTIMODAL COMPLEXITY

Data integration disconnect

Single optical images fail in clouds; optical, SAR & temporal passes remain uncoupled.

USER QUESTION

"What changed between these two passes?"

AGENTIC CONTROLLER

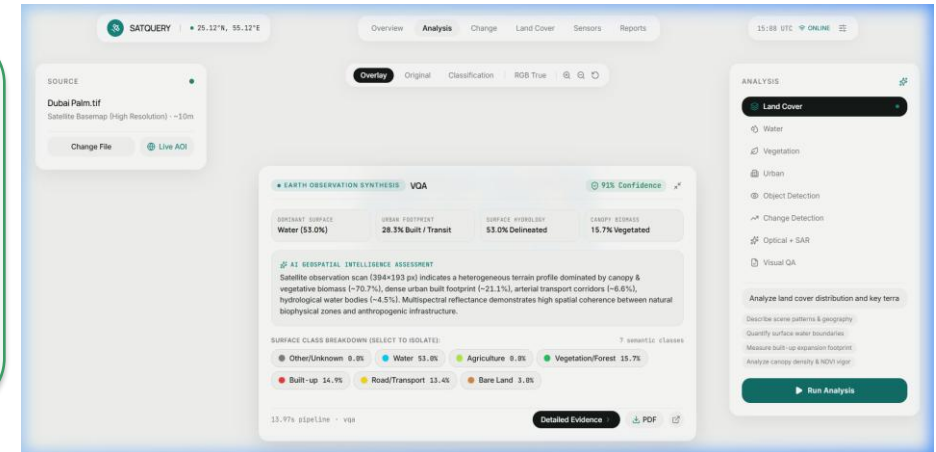
Autonomous intent & task classification

SPECIALIST AI

SegFormer · CDVQA
Optical + SAR Fusion

EVIDENCE OUTPUT

Pixel Masks · Hectares
Confidence & Audit Trace



01

NATURAL LANGUAGE

Plain text query replacing GIS commands

02

AGENTIC ROUTING

Autonomous model and tool selection

03

OPTICAL + SAR FUSION

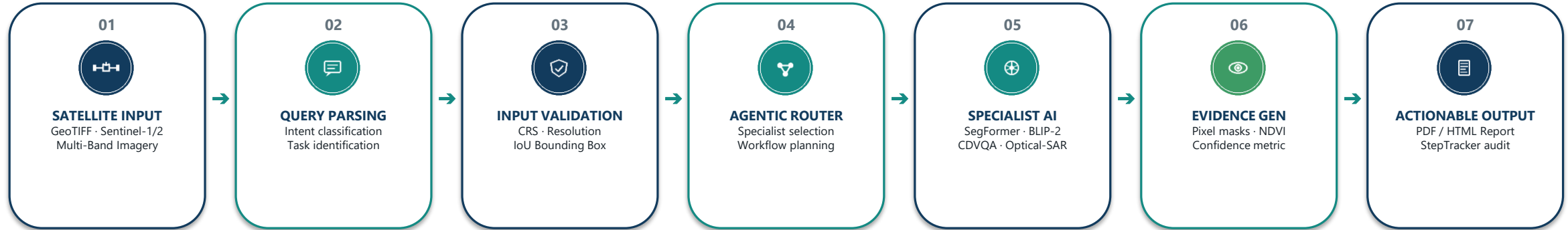
Multi-sensor day/night & cloud penetration

04

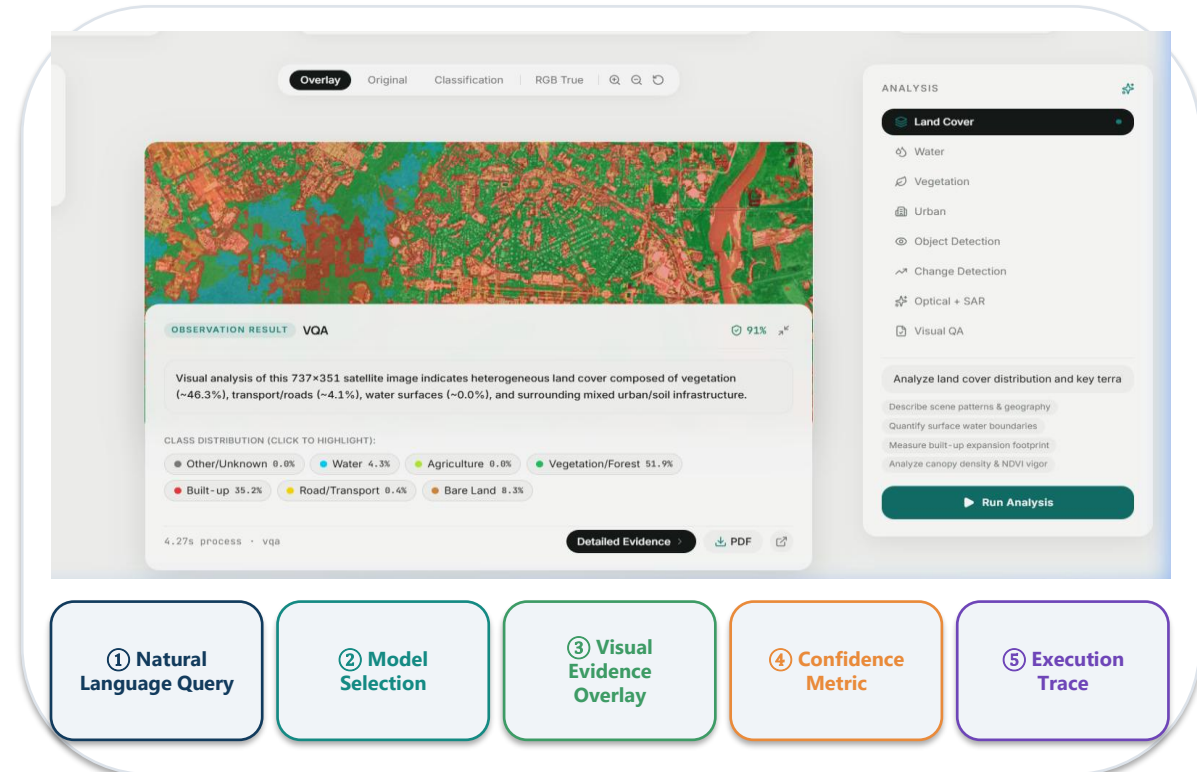
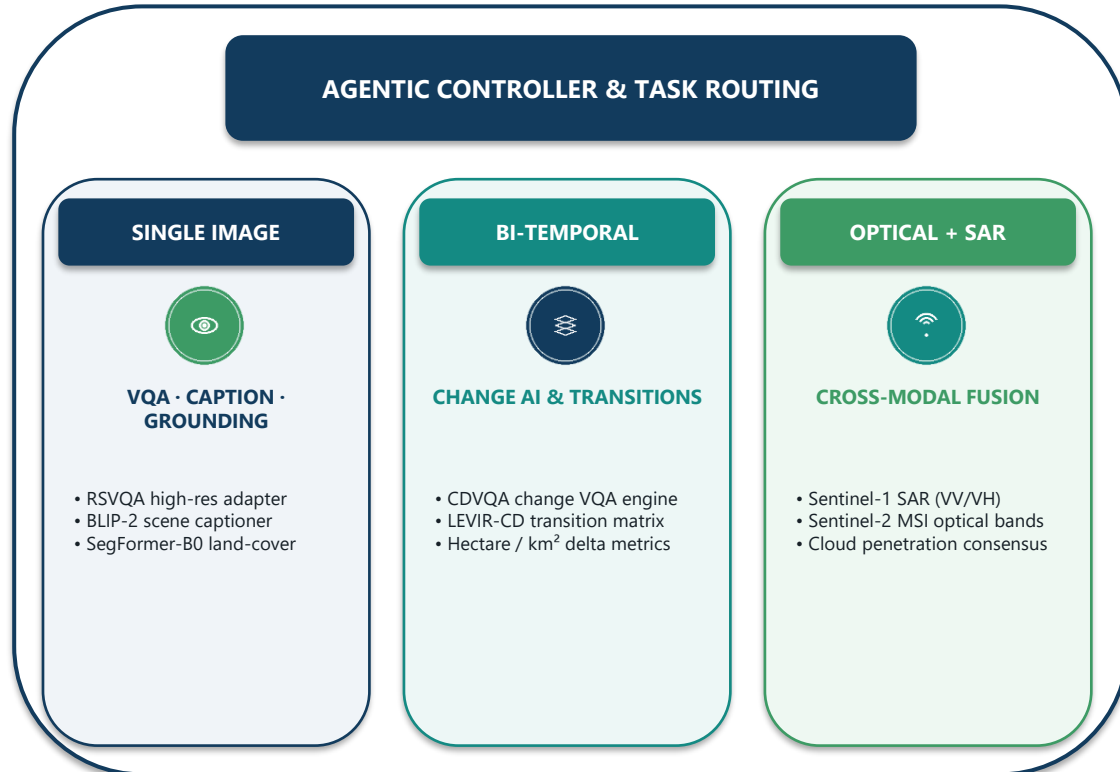
BI-TEMPORAL REASONING

Quantitative transition matrices & CDVQA

WHY SATQUERY? "One interface. Multiple specialist models. Evidence-grounded answers."



AGENTIC CONTROLLER & TASK ROUTING





TECHNICALLY FEASIBLE

**Open Geospatial Stack**

Rasterio, GDAL, EPSG:3857/4326 affine coordinate transforms.

**Open Satellite Datasets**

Sentinel-1/2 (Copernicus), Landsat-8/9, LandCover.ai benchmarks.

**Specialist AI Models**

SegFormer-B0, CDVQA, BLIP-2 with remote-sensing domain adapters.

**Web-Based Deployment**

FastAPI async pipeline + React Leaflet hardware-accelerated viewer.

POTENTIAL CHALLENGES

**Optical Image Quality**

Cloud occlusion, atmospheric haze, and night-time optical blindness.

**Optical-SAR Alignment**

Disparate sensor grids, spatial resolutions, and vantage angles.

**Large-Image Computation**

Multi-gigabyte GeoTIFF rasters strain memory and inference latency.

**AI Reliability & Trust**

Generic VLMs can hallucinate without spatial grounding verification.

MITIGATION STRATEGIES

**Complementary SAR Fusion**

SAR microwaves penetrate clouds and operate day/night unconditionally.

**Validate Spatial Compatibility**

InputValidator verifies CRS & IoU bounding box overlap before inference.

**Tiling & Patch Processing**

Sub-region patch extraction (512x512) and pyramid tiling prevent OOM.

**Evidence-Grounded Output**

Every response is tethered to pixel masks, spectral indices, & confidence.



BUILDABLE + SCALABLE + AUDITABLE: "Make complex multimodal satellite analysis accessible through one intelligent interface."



SUPPORTED MODALITIES

SINGLE · BI-TEMPORAL
OPTICAL + SAR

Optical MSI, SAR Radar (VV/VH), Multi-Temporal

SPECIALIST WORKFLOWS

21 REGISTERED TOOLS

VQA, SegFormer, CDVQA, Grounding, Spectral Indices

EVIDENCE OUTPUT

PIXEL MASKS & STATS

Exact Hectares, km², NDVI, NDWI, NDBI & Confidence

REPORT GENERATION

AUDIT-READY DOSSIERS

Dynamic ReportLab PDF & HTML + StepTracker Trace



AGRICULTURE

Crop & Vegetation Monitoring

Quantify crop vigor, drought stress & irrigation dynamics.
• 10m Sentinel-2 MSI • NDVI/EVI vigor tracking • Field stress alerts

WATER RESOURCES

Hydrology & Catchment Basins

Delineate surface water bodies, reservoirs & river morphology.
• Automated NDWI thresholding • Temporal surface retention delta

URBAN PLANNING

Built-Up Footprint & Zoning

Track urban sprawl, impervious surfaces & infrastructure growth.
• SegFormer semantic extraction • NDBI built-up indices • Zoning audit

FOREST & ECOLOGY

Biomass & Conservation

Surveil dense forest cover, illegal clearing & coastal buffers.
• Multi-temporal change delta • Coastal CRZ protection monitoring

DISASTER RESPONSE

All-Weather Flood & Hazards

Rapid damage assessment during cyclones, floods & landslides.
• Sentinel-1 C-band SAR • Pierces cloud cover & night • Hour-zero triage

EARTH OBSERVATION

Sovereign Intelligence & Access

Democratize complex EO rasters for non-GIS domain specialists.
• 21 registered specialist tools • WGS84 & GeoTIFF spatial integrity

SATELLITE DATA (Optical / SAR / Pairs) → AGENTIC AI → VISUAL EVIDENCE → ACTIONABLE INSIGHT → DECISION SUPPORT



RESEARCH FOUNDATIONS & BENCHMARKS

**BigEarthNet — Remote-Sensing Representation Learning**

Sumbul et al., IEEE IGARSS 2019 · Sentinel-2 multispectral band adaptation & land-cover taxonomy.

**RSVQA — Remote-Sensing Visual Question Answering**

Lobry et al., IEEE TGRS 2020 · Foundational formulation for spatial counting, presence & relation VQA.

**VRSBench — Vision-Language Grounding & Captioning**

Li et al., NeurIPS / IEEE 2024 · Evaluation benchmark for text-guided visual region retrieval.

**CDVQA — Bi-Temporal Change Detection VQA**

Yuan et al., IEEE GRSL 2022 · Informs change reasoning and quantitative transition matrix extraction.

**SegFormer — Semantic Transformer Segmentation**

Xie et al., NeurIPS 2021 · Core neural backbone powering 6-class land cover segmentation on LandCover.ai.

**SEN1-2 — Optical-SAR Multimodal Learning**

Schmitt et al., ISPRS Annals 2018 · Spatial co-registration & cross-spectral feature fusion standards.



HOW RESEARCH SUPPORTS SATQUERY

**BigEarthNet** → Remote Sensing Adaptation Layer**RSVQA** → Visual Question Answering (VQA) Module**VRSBench** → Region Grounding & Scene Captioning**CDVQA** → Bi-Temporal Change Detection Engine**SEN1-2** → Optical + SAR Cross-Modal Fusion**Indian Space Policy 2023** → Democratization of EO Data for Non-GIS Users

ISRO Bhuvan & NRSC Compatibility: Adheres to OGC standards, WGS84 (EPSG:4326), Web Mercator (EPSG:3857) and GeoTIFF specifications for interoperable national infrastructure.

RESEARCH → MODELS → AGENTIC ORCHESTRATION → EVIDENCE → APPLICATION